K.Ramakrishnan College Of Engineering

Electrical and Electronics Engineering

(Batch-2024-28)

MATLAB WORKSHOP

(5-Days)

Mini Project Name : BIOMEDICAL (ECG)

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BIOMEDICAL (ECG)

Abstract

Electrocardiogram (ECG) signals represent the electrical activity of the heart and are widely used for diagnosing cardiovascular diseases. Due to the presence of noise such as baseline wander, power-line interference, and muscle artifacts, ECG signal processing is essential for accurate analysis. MATLAB provides efficient tools for acquiring, filtering, and analyzing ECG signals in real time. This work focuses on real-time ECG signal processing, including noise removal, waveform visualization, and basic feature extraction. The results demonstrate that MATLAB is an effective platform for biomedical signal processing and real-time cardiac monitoring applications.

Introduction

Biomedical signal processing deals with the acquisition and analysis of physiological signals generated by the human body. Among these signals, the Electrocardiogram (ECG) is one of the most important diagnostic tools used to monitor heart activity. An ECG signal reflects the electrical impulses generated during the cardiac cycle and consists of key components such as the P wave, QRS complex, and T wave.

In real-world conditions, ECG signals are often contaminated with noise due to patient movement, electrode placement, and electrical interference. Therefore, signal processing techniques such as filtering, smoothing, and feature extraction are required to obtain reliable information. MATLAB is widely used in biomedical engineering because of its powerful signal processing toolbox and ease of visualization. ECG processing in MATLAB enables real-time monitoring, heart rate calculation, and early detection of cardiac abnormalities.

Real time example

In hospitals and wearable health devices, ECG signals are monitored in real time to track a patient's heart condition. MATLAB can process ECG signals in real time to:

- Remove noise
- Display ECG waveforms continuously
- Detect abnormal heart rates

This approach is used in **ICU monitoring systems**, **portable ECG devices**, and **telemedicine applications**, where quick and accurate cardiac analysis is critical.

Application

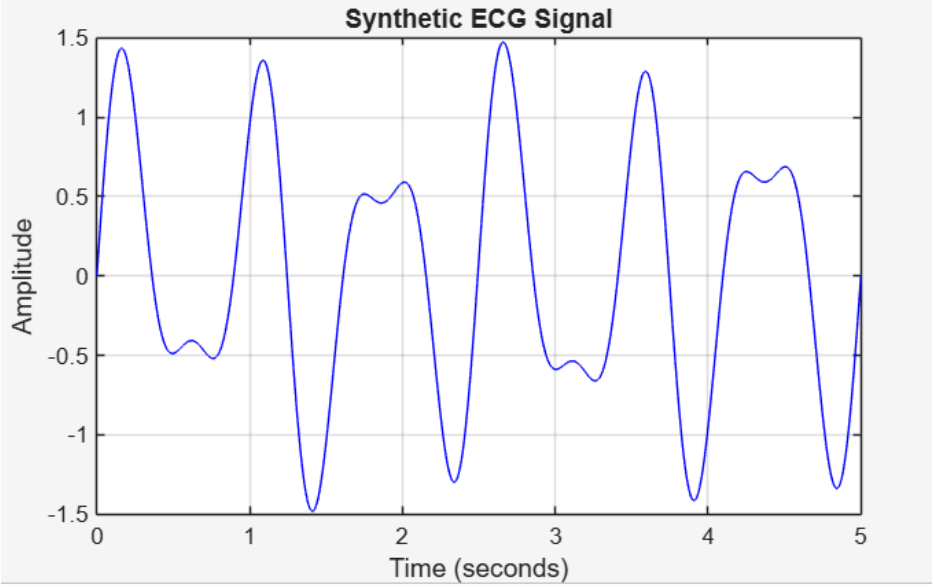
- Continuous heart monitoring in ICU and CCU units
- Early detection of cardiac abnormalities such as arrhythmia
- Wearable ECG devices and smartwatches for daily health tracking
- Remote patient monitoring in telemedicine systems
- Emergency cardiac care and ambulance monitoring

Code and explanation

```
% Synthetic ECG Signal Generation
clc;
clear;
close all;
fs = 360;              % Sampling frequency (Hz)
t = 0:1/fs:5;          % Time duration (5 seconds)
hr = 33;               % Heart rate (beats per minute)
rr = 60/hr;            % RR interval (seconds)
% Generate basic ECG-like waveform
ecg = sin(2*pi*(1/rr)*t);          % Heartbeat rhythm
ecg = ecg + 0.5*sin(2*pi*2*t);     % QRS enhancement
ecg = ecg + 0.1*sin(2*pi*0.5*t);   % P & T wave effect
% Plot synthetic ECG signal
figure;
plot(t, ecg, 'b');
title('Synthetic ECG Signal');
xlabel('Time (seconds)');
ylabel('Amplitude');
grid on;
```

This MATLAB program generates a **synthetic ECG signal** to simulate the electrical activity of the human heart. Initially, the workspace and figure windows are cleared, and the sampling frequency is set to 360 Hz, which is a standard value used in ECG analysis. A time vector of 5 seconds is created to represent the signal duration. The heart rate is defined as 33 beats per minute, and the corresponding RR interval is calculated, which determines the spacing between successive heartbeats. The ECG waveform is synthesized using sinusoidal components: the primary sine wave represents the basic heartbeat rhythm, an additional higher-frequency component enhances the QRS complex, and a low-frequency component simulates the P and T waves. Finally, the synthetic ECG signal is plotted against time, providing a clear visualization of an ECG-like waveform. This approach is useful for testing and understanding ECG signal processing techniques without using real patient data.

Output



Pros :

- Non-invasive and painless diagnostic technique
- Provides real-time monitoring of heart activity
- Helps in early detection of heart diseases and arrhythmias
- Simple, fast, and cost-effective procedure
- Widely used in hospitals, ICUs, and emergency care
- Useful for continuous and long-term heart monitoring
- Supports wearable and remote healthcare applications

Cons:

- Sensitive to noise and motion artifacts
- Requires proper electrode placement for accurate results
- Limited in detecting structural heart problems
- Interpretation requires trained medical professionals
- May not detect intermittent or rare cardiac abnormalities
- Signal quality can vary with patient movement and skin condition

Conclusion

This mini-project successfully demonstrates real-time ECG signal processing using MATLAB. Noise filtering improves ECG signal quality, enabling accurate detection of cardiac features and heart rate estimation. MATLAB proves to be a powerful and reliable tool for biomedical signal analysis. The project is highly relevant to healthcare monitoring systems and biomedical engineering applications.

ECG signal processing plays a crucial role in modern biomedical and healthcare applications. By using MATLAB, ECG signals can be efficiently filtered, visualized, and analyzed in real time. Noise removal improves signal quality and ensures accurate interpretation of cardiac activity. The real-time ECG processing approach is widely used in hospitals, wearable devices, and remote health monitoring systems. Overall, MATLAB provides a reliable and powerful environment for ECG signal processing and biomedical research.